1.

$$2^9 = 512$$

$$\boxed{512}$$

2.

$$(110101.011)_2 = 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3}$$

$$= 32 + 16 + 4 + 1 + \frac{1}{4} + \frac{1}{8} = 53.375$$

$$\boxed{(110101.011)_2 = (53.375)_{10}}$$

3.

$$\begin{aligned}
912 \div 2 &= 456 \text{ R } 0 \\
456 \div 2 &= 228 \text{ R } 0 \\
228 \div 2 &= 114 \text{ R } 0 \\
114 \div 2 &= 57 \text{ R } 0 \\
57 \div 2 &= 28 \text{ R } 1 \\
28 \div 2 &= 14 \text{ R } 0 \\
14 \div 2 &= 7 \text{ R } 0 \\
7 \div 2 &= 3 \text{ R } 1 \\
3 \div 2 &= 1 \text{ R } 1 \\
1 \div 2 &= 0 \text{ R } 1
\end{aligned}$$

Reading the remainders from bottom to top gives

$$\boxed{(912)_{10} = (1110010000)_2}.$$

4.

$$\begin{aligned}
845 \div 16 &= 52 \text{ R } 13 = D \\
52 \div 16 &= 3 \text{ R } 4 \\
3 \div 16 &= 0 \text{ R } 3
\end{aligned}$$

$$\boxed{(845)_{10} = (34D)_{16}}$$

5.

$$6 \to 0110, \qquad C \to 1100, \qquad 5 \to 0101, \qquad A \to 1010$$

$$\boxed{(6C5.A)_{16} = (0110\ 1100\ 0101.1010)_2}$$

6.

$$(111010110.101)_2 = (111\ 010\ 110.101)_2$$

$$111 \to 7, \qquad 010 \to 2, \qquad 110 \to 6, \qquad 101 \to 5$$

$$\boxed{(111010110.101)_2 = (726.5)_8}$$

7.

$$\begin{array}{r}
1110110 \\
+0101101 \\
\hline
10100011
\end{array}$$

$$\boxed{(1110110)_2 + (101101)_2 = (10100011)_2}$$

8.

$$\begin{array}{r} 10110 \\ 101100 \\ 0000000 \\ +10110000 \\ \hline 11110010 \end{array}$$

$$\boxed{(10110)_2 \times (1011)_2 = (11110010)_2}$$

9.

$$+37 = (00100101)_2, \qquad +62 = (00111110)_2$$

$$-62: \quad 00111110 \to 11000001 \to 11000010$$

$$\begin{array}{r} 00100101 \\ +11000010 \\ \hline 11100111 \end{array}$$

The sign bit is 1, so the result is negative. Its magnitude is

$$11100111 \to 00011000 \to 00011001 = (25)_{10}.$$

$$\boxed{(+37) - (+62) = -25}$$

10.

$$384 = 0011\ 1000\ 0100, \qquad 529 = 0101\ 0010\ 1001$$

Ones:

$$0100 + 1001 = 1101, \quad 1101 + 0110 = 1\ 0011$$

Tens:

$$1000 + 0010 + 1 = 1011, \quad 1011 + 0110 = 1\ 0001$$

Hundreds:

$$0011 + 0101 + 1 = 1001$$

$$\boxed{384 + 529 = 1001\ 0001\ 0011 \text{ in BCD } = 913}$$